

Financial Stability and Market Liquidity

Learning Objectives:

- The provision of liquidity by market makers and arbitrageurs hinges on their access to funding liquidity
- Funding and market liquidity can feed on each other, generating “liquidity spirals”
- This feedback loop can generate financial instability
- Interconnections between financial intermediaries and markets can propagate instability and generate systemic risk
- Adverse selection due to counterparty risk contributes to instability
- Public provision of funding liquidity can mitigate such instability

11.1 Introduction

Liquidity fluctuates over time, to the point that sometimes it evaporates completely, especially in the context of financial crises. This chapter considers the causes of these sharp drops in liquidity and their effects on asset prices, and the channels by which they can propagate across markets. We shall see that the provision of market liquidity crucially depends on the financial health of the intermediaries who provide liquidity services, and in particular on their access to funding.

These intermediaries not only include the market makers described in Chapter 3, who speedily absorb the orders placed by noise traders into their inventories, but also professional investors such as hedge fund and mutual fund managers, who play the same function at lower frequencies (ranging from hours to weeks). For instance, when a large flow of sell orders placed by uninformed traders tends to push the market price below the fundamental, these professional investors “lean against the wind” by absorbing these sell orders in their portfolios; conversely, in response to large buy orders from noise traders, they tend to sell their holdings or even short-sell the asset.

Hedge funds that follow “contrarian” strategies for extended periods of time are good examples. They build portfolios with long positions in underperforming stocks and short positions in overperforming stocks, so as to exploit the price reversals that should occur if the price movements are due to transient noise trading shocks. Like traditional market makers, they sustain inventory risks because the extent and the speed of price reversals are uncertain, although the price reversals that they exploit take place at a lower frequency than those on which traditional market makers thrive. And, just as with market makers, their activity tends to mitigate mispricing, that is, the deviation of the market price from fundamentals that noise trading would otherwise generate, or to reduce its persistence. By the same token, their activity helps reassert the “law of one price,” which requires assets with the same cash flows to trade at the same price: large orders placed by uninformed traders may otherwise cause prices to violate this non-arbitrage condition. For this reason, these professional speculators are sometimes referred to as “arbitrageurs.”

This chapter highlights two novel points regarding the ability of market makers and arbitrageurs to provide market liquidity and prevent persistent mispricing. First, Section 11.2 shows that their ability to provide market liquidity hinges on their access to credit, hence on “funding liquidity.” To see why, notice that to absorb large sell orders that exceed the cash in their portfolios, market makers and arbitrageurs must be able to borrow from banks or other financial intermediaries; similarly, to absorb large buy orders that exceed their holdings of the relevant asset, they must be able to short-sell it, and to fund such short sales they may again need to borrow the necessary resources. The need for such funding liquidity is particularly important when the orders generated by noise traders are correlated (Section 11.3).

Second, the availability of “funding liquidity” may itself be affected by dislocations in asset prices, as shown in Section 11.4: the mispricing generated by a bout of noise trading may become so extreme and persistent as to generate a drop in funding liquidity. For instance, if market makers and arbitrageurs have taken long positions in stocks that they consider undervalued, and are booking losses on these positions because stock sales by noise traders persist, the banks that fund these positions may refuse to roll over their loans to them. This is because the drop in stock prices may inflict capital losses on banks, either due to the implied increase in their credit risk exposure to market makers and arbitrageurs or due to banks’ own exposure to the mispriced stocks: compliance with regulatory capital ratios requires them to reduce lending in line with their reduced capital. Hence, there can be a feedback effect from market liquidity to funding liquidity.

The mutual interaction between market and funding liquidity just described may generate a perverse feedback loop, as illustrated in Figure 11.1 for the case

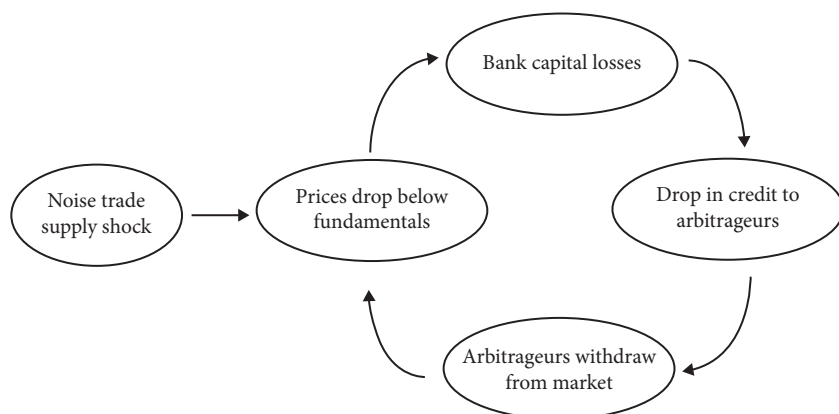


Figure 11.1 Feedback loop between market liquidity and funding liquidity.

where the initial trigger is a large flow of sell orders placed by noise traders. If the market is illiquid, these sales will push asset prices below fundamentals, and if this mispricing is large enough, it may induce a drop in funding liquidity, which in turn impairs the ability of market makers and arbitrageurs' ability to absorb orders, precipitating extreme mispricing and capital losses for banks.

The feedback loop illustrated in this figure can threaten the stability of financial markets, not only because of the magnitude of the asset price changes that they may trigger, but also because these impair simultaneously the balance sheets of market makers, arbitrageurs (e.g., hedge funds) and banks. Insofar as these financial intermediaries are typically interconnected via a network of contractual relationships in different markets, these destabilizing effects can propagate across intermediaries via their interlocking balance sheets, as discussed in Section 11.5: the liabilities of each are the assets of others. In extreme situations, the resulting domino effect can threaten the stability of the whole financial system. Hence, this chapter links market liquidity not only to funding liquidity, but also to the notion of "systemic risk," i.e., the risk that swings in asset prices and in market liquidity may result in a financial crisis.

The chapter closes with Section 11.6, which highlights that the balance sheets of financial intermediaries matter not only for market liquidity because of their role in amplifying and transmitting asset price shocks, but also because they can be a source of adverse selection. Recall that in Chapters 3 and 4 of this book private information about asset fundamentals has been shown to generate adverse selection and thereby market illiquidity. But the same can be said of private information about the balance sheets of financial intermediaries. If there are doubts about the ability of intermediaries to fulfill their obligations, this will generate adverse selection in the markets where they operate. This is relevant

for counterparty risk in security markets, as well as in the interbank market, where banks are the issuers of the credit claims being traded, and in structured debt markets where the quality of the asset is affected by the standing of an intermediary. For instance, the value of an asset-backed security (ABS) that is partly guaranteed by a bank depends on the creditworthiness of the bank itself; and the other financial intermediaries typically do not have as accurate and timely a gauge of that as the bank in question.¹ At times of financial stress, creditworthiness typically becomes much more important than in normal times, and therefore the implied adverse selection becomes a more serious issue for the liquidity of the relevant assets, as witnessed by the dry-up of ABS markets during the financial crisis of 2008. Hence, private information about the state of intermediaries' balance sheets can be an additional threat to systemic stability.

11.2 Funding Liquidity and Limits to Arbitrage

The absence of arbitrage is a central tenet of asset pricing: two assets with identical cash flows should command the same price. Yet there are well-known cases in which this principle appears to be violated, due to frictions that prevent arbitrage between these assets. Transaction costs, for instance, may exceed the profits that can be gained by exploiting a price discrepancy, so that arbitrageurs will have no incentive to eliminate such discrepancy. In this section we focus on a less obvious source of frictions, namely the availability of funding for arbitrage.

One example of persistent deviations from no-arbitrage prices is the failure of the covered interest rate parity (CIP) condition in the foreign exchange market. This condition ensures that investors earn the same return from debt denominated in domestic and foreign currency, if they can rely on forward contracts to eliminate foreign exchange risk. It states that the interest rate differential between the two currencies in the respective cash money markets should equal the differential between the forward and spot exchange rates. The fact that this condition does not always hold opens up potential opportunities to earn riskless profits from covered interest arbitrage. Du, Tepper, and Verdelhan (2018) document that the CIP condition is systematically and persistently violated among G10 currencies, leading to significant arbitrage opportunities in currency and fixed income markets since the 2008 global financial crisis, and

¹ ABS are securities backed by a portfolio of income-generating assets such as housing mortgage loans, student or car loans, or credit card receivables. Examples of ABS are mortgage-backed securities (MBS) and collateralized debt obligations (CDO). Investors who purchase these securities receive the cash flow generated by the underlying loan pool. The bank that designs and sells the ABS to investors may provide guarantees to the investors who purchase the ABS to reduce the risk stemming from the possible default of the loans included in its portfolio.

document that these deviations for major currencies are not explained away by credit risk or transaction costs. This is striking, the foreign exchange forward and swap market being among the largest and most liquid derivative markets in the world.

Another instance of persistent arbitrage opportunities is the opening example in Section 10.2.1: in the United States, Treasury notes typically trade at a discount relative to otherwise identical Treasury bills. We argued that this difference in gross rates of return can be accounted for by the greater liquidity of bills. But why don't arbitrageurs eliminate this difference by selling bills, and buying notes, holding them to maturity so as to avoid further trading costs? This looks like a textbook example of a profitable arbitrage opportunity—the kind that according to asset pricing theory cannot last.

But lack of funding can prevent arbitrageurs from taking such an opportunity. In theory, riskless arbitrage should not require arbitrageurs to invest their own wealth or obtain outside funding, because a long position can be funded by an offsetting short position: arbitrage should be self-funding! However, in practice arbitrage does require capital. To understand this point, consider our simple example. It looks as if no capital is required to build the arbitrage portfolio: if the notes are worth 95% of the bills with the same maturity, the arbitrageur can use the proceeds from selling the bills short (say, \$100,000) to buy the notes (\$95,000), and immediately cash in the difference (\$5,000), which is an instantaneous arbitrage profit (a “free lunch”), provided he can hold the position to maturity.

In reality, though, this kind of standard arbitrage portfolio requires capital. For instance, to sell bills short an arbitrageur must borrow them from a broker, who for example may require a margin of 150% of the value of the short sale as collateral (in our example, \$150,000); buying the notes will cost another \$95,000.² Therefore, in this example the total capital, net of the proceeds of the short sale, committed to the arbitrage portfolio would be \$145,000, a huge multiple of the arbitrage profits (\$5,000).

Hence a reason for the persistence of arbitrage opportunities is that arbitrageurs may not have enough wealth to exploit them. Of course they could borrow so as to carry out the arbitrage. But unless they can secure long-term funding, borrowing exposes them to the risk of having to liquidate their position

² Under Regulation T, the Federal Reserve Board requires all short sale accounts to have 150% of the value of the short sale at the time the sale is initiated. The sale described in the text is a “covered short sale,” where the seller borrows the stock needed to deliver it to the buyer. In principle, even “naked shorting” may be allowed, whereby the seller does not borrow the stock needed and therefore must buy it on the market for delivery, which creates the risk of failure to deliver the stock to the buyer. However, in practice “naked shorting” has been banned in most jurisdictions since the financial crisis of 2008.

before the prices of notes and bills eventually converge at maturity. This risk is particularly great if the mispricing that they try to correct widens: as a result, they may have to meet larger margin requirements or may face a cut in funding due to a loss of confidence in their strategy. So, paradoxically, they may be forced to liquidate their position just when its future profitability is largest.

Limited wealth generally forces arbitrageurs to raise external funds. If these funds come in the form of short-maturity debt or can be withdrawn at will, arbitrageurs risk being unable to refinance their positions before their arbitrage portfolio pays off. For instance, in our example, the lender of the bills may recall them at any point in time, forcing the arbitrageur to either borrow them from another broker or to close out his position, possibly at a loss. Indeed, Du et al. (2018) find that deviations from CIP increase at the end of each quarter, especially for contracts that appear in banks' balance sheets, when banks face tighter capital constraints due to quarterly regulatory filings.³

One way out of these problems is to obtain long-term financing, as some arbitrageurs do by imposing lock-up covenants, the condition that investors cannot withdraw their funds for some minimum period. For instance, hedge funds typically raise money in the form of shares that can be redeemed at their market value only after a lock-up period. However, if the claims are short-term, financiers can better control the use of their funds by arbitrageurs: if the latter do not perform well, external investors can simply pull the plug. This threat helps to discipline arbitrageurs and prevent excessive risk taking. Alternatively, short-term financing could be an optimal response to uncertainty: it allows the financier to progressively increase his stake as he becomes more confident about the real ability of arbitrageurs. But, as noted above, it exposes arbitrageurs to the danger of early liquidation. The demise of LTCM, a major U.S. hedge fund, in 1998 (see Box 11.1) offers a vivid illustration.

This reasoning has been used to explain the empirical evidence of persistent arbitrage opportunities. For instance, Rosenthal and Young (1990), Froot and Dabora (1999), and de Jong, Rosenthal, and Van Dijk (2009) show that the prices of dually listed stocks (so-called twin stocks) often differ even though they are claims on the same cash flows.⁴ De Jong, Rosenthal, and Van Dijk (2009) show that arbitrageurs who try to exploit these price differences must hold their positions for long periods of time with all the attendant risk. They write: "Since

³ Note that arbitrage can also be performed by a bank's own security trading desk, with funding provided by the bank itself. Even so, its security trading activity is constrained by the bank's capital requirements.

⁴ A "dually listed company" is a corporate structure that involves two listed companies with two distinct stocks that are claims on the same assets. A "cross-listed" company is a single firm with a single stock traded in multiple markets; researchers have found that arbitrage opportunities on these stocks are much smaller and less common than for the dually listed.

there is no identifiable date at which dual listed companies prices will converge, arbitrageurs with limited horizons who are unable to close the price gap on their own face considerable uncertainty. In some cases, arbitrageurs would have to wait for almost nine years before prices have converged and the position is closed. In the short run, the mispricing might deepen. In these situations, arbitrageurs receive margin calls, after which they would most likely be forced to liquidate part of the position at a highly unfavorable moment and suffer a loss” (p. 497).

Box 11.1 The LTCM Crisis

Long-Term Capital Management (LTCM) was a hedge fund founded in 1994 with equity of \$1.3 billion. Investors had to put up a minimum of \$1 million and could not withdraw it for three years. The first four years of its life were extremely profitable. Its main investment was of “market-neutral arbitrage,” i.e., arbitrage positions that involve no systematic risk, mainly buying high-yield illiquid bonds (e.g., emerging market bonds and non-investment-grade corporate bonds) and shorting low-yield liquid bonds (e.g., U.S. Treasuries). Many these arbitrage positions were taken indirectly via derivative contracts such as rate swaps. Thus LTCM was betting that the yield spread between low- and high-risk bonds would narrow. Even by the standards of hedge funds, LTCM’s leverage was enormous (over \$20 of borrowing per \$1 of equity), so that even a slight narrowing of yield spreads would translate into huge profits.

In the spring of 1998, however, spreads widened in the wake of the Southeast Asian financial collapse and Russia’s troubles. The situation deteriorated further in August, when Russia devalued the ruble and declared a moratorium on public debt payments. By September, the yield spread on the J.P. Morgan index of emerging market bonds had increased to over seventeen percentage points, five times more than in October 1997. Yield spreads on U.S. non-investment-grade bonds also rose fivefold over the period. As a result, LTCM started making huge losses, cutting its equity value to \$600 million by September (from \$4.8 billion at the start of 1998). At this point, both LTCM and other funds with similar strategies came under pressure to meet margin calls and post additional collateral with creditors and swap counterparties, resulting in pressure to liquidate their positions at a further loss.

However, in fear that a generalized flight to quality might trigger a downward price spiral and a cascade of bankruptcies of financial institutions, the Federal Reserve persuaded a consortium of LTCM’s major creditors to inject over \$3.6 billion into the fund in return for a 90% equity stake. After the

rescue, LTCM was run by these creditors and, helped by the recovery of financial markets, it gained 13% by December 1998. In the following months its portfolio was unwound and LTCM was liquidated.

The sequence of events in the LTCM crisis offers a good illustration of the forces at work in the model of Section 11.2.1: LTCM was arbitraging away yield differences that it considered excessive, but in doing so it was exposed to the risk of early liquidation by its creditors. Its positions were fundamentally sound, in that its portfolio bounced back once its creditors were persuaded to extend long-term financing, but the differences were not eliminated as quickly as LTCM had bet they would be. Other arbitrageurs did not intervene on a sufficient scale, because the Russian and Asian crises had raised volatility and tightened the funding available for arbitrage activity.

11.2.1 Risk of Early Liquidation as a Limit to Arbitrage

To illustrate these points, consider the following simple model, which is inspired by Shleifer and Vishny (1997). There are three dates (0, 1, and 2) and two zero-coupon bonds (A and B) with the same certain payoff V at date 2. The risk-free interest rate is set to zero, so that absent arbitrage opportunities, the price of the two bonds should equal V at both dates 0 and 1.

Suppose that an arbitrage opportunity arises at date 0: the price of bond A falls below that of bond B by an amount $M_0 > 0$ ($P_{0A} = P_{0B} - M_0$), while bond B is priced correctly ($P_{0B} = V$). That is, bond A is undervalued, and the size of this mispricing at date 0 is M_0 . At date 1, the mispricing increases to $M_1 > M_0$ with probability κ or disappears with probability $1 - \kappa$. In the first case, the price of bond A is $P_{1A} = P_{1B} - M_1$; in the second case it is simply $P_{1A} = P_{1B} = V$. In the next section, we explain why such mispricing may arise and persist. At date 2, the assets pay off so that any mispricing is eliminated: $P_{2A} = P_{2B} = V$ with certainty.

As the prices of bonds A and B will converge at date $t = 2$, the arbitrage portfolio is a short position in asset B and an offsetting long position in asset A .⁵ We assume that arbitrageurs must choose whether to build their arbitrage portfolio at date 0 or at date 1, denoting the decision to intervene at date t by the indicator function $I_t \in \{0, 1\}$: if $I_t = 1$, the arbitrageur sells one share of B and buys one share of A at date t . If $I_0 = 1$, the portfolio is built at date 0; if $I_1 = 1$, at date 1. Otherwise the arbitrageur is inactive at both dates ($I_0 = I_1 = 0$). There are two restrictions: the arbitrageur cannot be active in both periods, that

⁵ This type of arbitrage which bets on the convergence in the prices of two related securities, is known as "convergence trading."

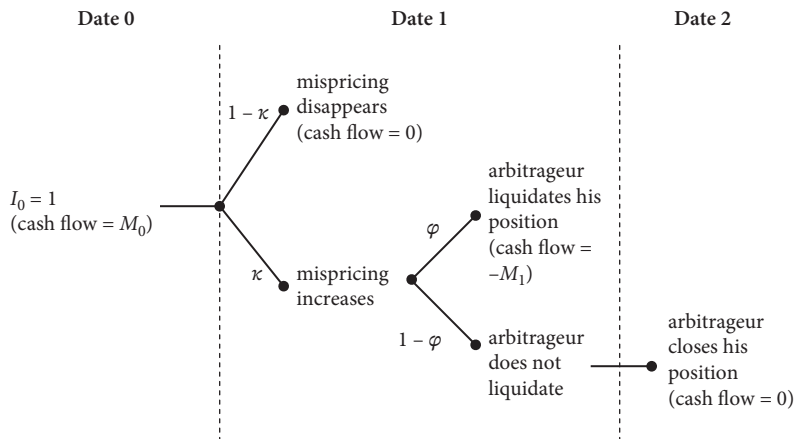


Figure 11.2 Actions and cash flows if the arbitrageur intervenes at date 0.

is, he cannot choose $I_0 = I_1 = 1$, and cannot take a position of more than one unit in the two bonds. These assumptions are intended to capture the funding constraints discussed above.

Suppose that the arbitrageur chooses to intervene at date 0. In this case, Figure 11.2 shows the cash flows of his portfolio at each date and in each contingency. The arbitrageur pockets the mispricing M_0 at date 0. With probability $1 - \kappa$ the mispricing disappears at date 1, so he closes his position. But with probability κ the mispricing increases to M_1 .

When the mispricing persists at date 1, outside investors may choose to cut their funding of the arbitrageur's activities because they cannot tell whether it is due to a deepening of the arbitrage opportunity or a miscalculation by the arbitrageur. Then the arbitrageur must close his position at a loss: so the model captures the notion of "performance-based arbitrage," namely, that funds under management are withdrawn from arbitrageurs following trading losses. We assume that such forced liquidation occurs with probability φ . When forced to liquidate, the arbitrageur has a negative cash flow of $-M_1$, because he resells security A at price $V - M_1$ and covers his short position by buying security B at price V . If he is not forced to liquidate at date 1, he has zero cash flow at $t = 1$ and $t = 2$, as at this date the net payoff of his arbitrage portfolio is zero. Hence, intervention at date 0 yields an expected profit equal to $\Pi_0(\varphi) = M_0 - \kappa\varphi M_1$.

If instead the arbitrageur has not intervened at date 0, he will intervene at date 1 if the mispricing persists. The cash flows of his arbitrage portfolio in this case are illustrated in Figure 11.3. By definition he has zero cash flows at date 0. At date 1, he has the opportunity to set up an arbitrage portfolio only if the mispricing of bond A increases: hence he gets a cash flow M_1 with probability κ .

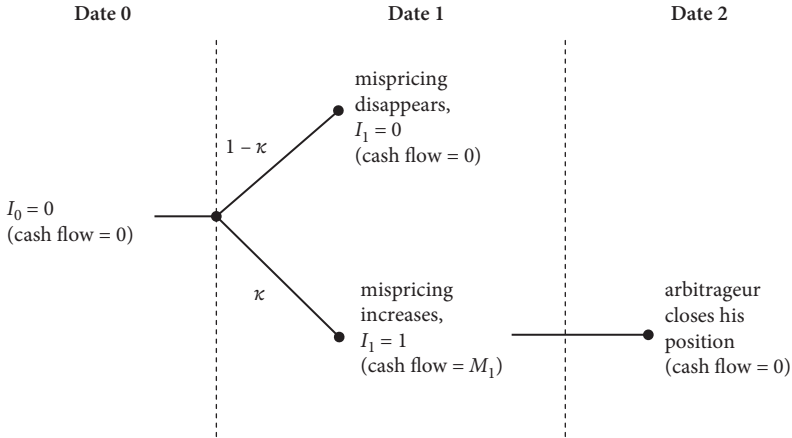


Figure 11.3 Actions and cash flows if the arbitrageur intervenes at date 1.

If instead the mispricing disappears, the arbitrageur can no longer intervene profitably, and will get no cash flow. Thus, the arbitrageur's expected profit is $\Pi_1 = \kappa M_1$.⁶

The choice hinges on the following trade-off. If the arbitrageur chooses to "wait" and intervene at date 1, he expects a gain M_1 if the mispricing persists (which happens with probability κ), and avoids the risk of early liquidation. But he forgoes the gain M_0 from exploiting the mispricing at date 0. Therefore, if $\kappa M_1 \geq M_0$, all arbitrageurs prefer to defer their intervention, as on average the mispricing will be greater at date 1. Hence, from now on, we focus on the more interesting case in which $\kappa M_1 < M_0$, that is, mispricing is expected to decrease. In this case, waiting is optimal if and only if $\Pi_1 > \Pi_0$, that is, if $\kappa M_1 > M_0 - \kappa \varphi M_1$. Hence, waiting is preferable if the future expected future mispricing κM_1 is large enough relative to the current mispricing M_0 and/or if the probability of early liquidation φ is large enough.

The threshold value of mispricing at $t = 1$ that keeps investors just indifferent between intervening early or late is:

$$\hat{M}_1 = \frac{M_0}{\kappa(1 + \varphi)} \quad (11.1)$$

The critical level of future mispricing $\hat{M}_1$ that makes arbitrageurs indifferent between intervening at date 0 or 1, is decreasing in the risk φ of early liquidation,

⁶ One can imagine a more complex setting in which an arbitrageur is allowed to invest only part of his resources at date 0 and "hoard" the rest to intervene at date 1. This case is more realistic but does not deliver additional insights.

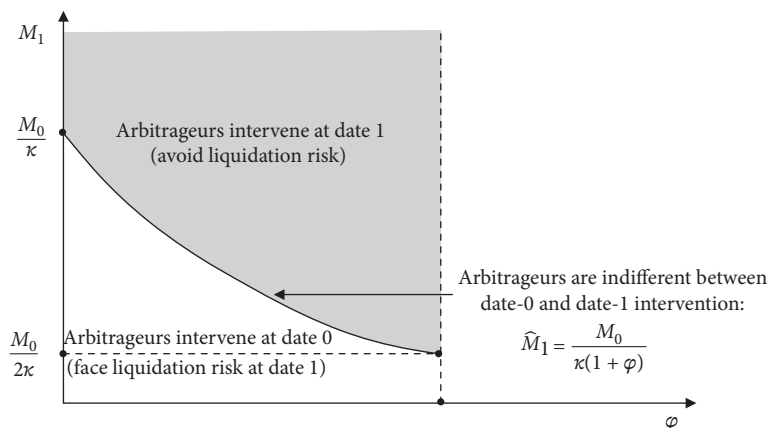


Figure 11.4 Allocation of arbitrage capital between dates 0 and 1.

as shown in Figure 11.4. In the shaded region above the “indifference curve” future mispricing and/or liquidation risk are large, so that arbitrageurs prefer to defer intervention. Below it, future mispricing and/or liquidation risk are low enough that they intervene immediately.

If expected mispricing were to exceed the level in expression (11.1), i.e., $M_1 > \hat{M}_1$, then all investors would choose to keep their powder dry and intervene at date 1.⁷ Conversely, if the opposite inequality $M_1 < \hat{M}_1$ were to hold, then all investors would intervene at date 0. Finally, if $M_1 = \hat{M}_1$, then all investors would be indifferent between intervening at date 0 and doing so at date 1. In this case, we assume that a fraction $f \in (0, 1)$ of investors intervenes at date 0 and a complementary fraction $1 - f$ keeps their powder dry for speculation at date 1. The fraction f is determined by market clearing at date 1, as will be seen in the next subsection.

11.2.2 Limited Speculative Capital as a Barrier to Arbitrage

So far, we have considered mispricing (M_0 and M_1) at either date as exogenous. The literature on limits to arbitrage often assumes that mispricing arises from supply or demand shocks that drive prices away from fundamental values, which may occur for various reasons. For instance, some investors (“noise traders”) may become irrationally over-pessimistic or over-optimistic (investor

⁷ Note that, even when mispricing is expected to decline ($\kappa M_1 < M_0$), arbitrageurs will prefer to postpone intervention to date 1 if the risk φ of early liquidation is large enough, that is, it can still be that $M_1 > \hat{M}_1$.

sentiment) about the prospects of a security.⁸ Alternatively, some investors (e.g., mutual funds or insurance companies) may suddenly be forced to liquidate positions in financial assets to deal with an unexpected funding need (e.g., to pay back creditors or meet investor withdrawals) or a prudential regulation requiring them to scale back their positions in certain asset classes. Such forced asset liquidations, which typically occur at deeply discounted prices, are known as “fire sales.”⁹

To determine in the simplest possible way the mispricing M_1 at date 1, suppose that it arises from a shock to noise traders’ supply—a “fire sale”—that occurs with probability κ and pushes the price of bond A further below its fundamental value. Otherwise, the price of bond A reverts to its fundamental value V . When this shock occurs, the aggregate noise traders’ supply of bond A at date 1, $y(P_{A1})$, is assumed to be:

$$y(P_{A1}) = 1 + \delta(P_{A1} - V) = 1 - \delta M_1, \quad (11.2)$$

where $\delta > 0$ is the sensitivity of the noise traders’ supply to the asset price. Hence, with probability κ , noise traders sell the bond at any price above $V - \frac{1}{\delta}$: they are overly pessimistic, as they value the bond at $V - \frac{1}{\delta}$, even though it will pay V for sure at date 2. Their sell orders exert a downward price pressure as long as the price is higher than $V - \frac{1}{\delta}$, that is, provided its undervaluation M_1 does not exceed $\frac{1}{\delta}$.

Hence, noise traders’ orders are a source of mispricing, which can be corrected only if arbitrageurs lean strongly enough against them. However, the arbitrageurs’ ability to do this is limited, since they must allocate capital between different strategies (in our model, interventions at date 0 and at date 1). Suppose that mispricing at date 0 is very pronounced. In this case, arbitrage capital will be massively invested to harness the date 0 mispricing, leaving little available to bet against mispricing at date 1. Mispricing will tend to persist and be increasing in its initial value, M_0 .

Thus the precise level of mispricing prevailing on the market if noise traders materialize at date 1 depends on the fraction f of arbitrageurs that intervened at date 0: at date 1, these investors are no longer available to speculate against noise traders’ sales, and a fraction ϕ of them will actually exacerbate the depressing effect of noise traders’ supply on the price, since these arbitrageurs will have to liquidate their positions prematurely. Note that ex ante ϕ is the probability that

⁸ Hence, the literature on limits to arbitrage is also related to that on behavioral finance, which studies how investors’ psychological biases affect securities prices and can explain market anomalies (such as the existence of arbitrage opportunities).

⁹ Shleifer and Vishny (2011) provides examples of fire sales and a review of the relevant literature.

each arbitrageur attaches to early liquidation, and ex post it is the fraction of arbitrageurs who face early liquidation at date 1.

The mispricing M_1 is determined by market clearing at date $t = 1$ when a noise trader supply shock occurs. The equilibrium price of bond A must be such that its supply is equal to its demand. There are two types of sellers: (i) noise traders who collectively sell an amount $y(P_{A1})$, which is positive for $M_1 < \frac{1}{\delta}$, and (ii) arbitrageurs who are forced to liquidate their position prematurely. As each arbitrageur who intervened at date 0 must liquidate his position with probability φ , the aggregate supply from arbitrageurs who liquidate at date $t = 1$ is $f\varphi$. Sell orders from these two categories of investors must be absorbed by the $1 - f$ arbitrageurs who still have capital to invest at date 1 because they did not intervene at date 0. Hence, the equilibrium price of asset A at date $t = 1$ is given by the condition:

$$\underbrace{y(P_{A1})}_{\text{noise traders' sell orders}} + \underbrace{f\varphi}_{\text{arbitrageurs' sell orders}} = \underbrace{1-f}_{\text{arbitrageurs' buy orders}} \quad (11.3)$$

Replacing $y(P_{A1})$ with its expression in (11.2), this market clearing condition yields the following expression for the equilibrium mispricing of bond A at date $t = 1$ if noise traders materialize (which they do with probability κ):

$$M_1 = \frac{1+\varphi}{\delta}f. \quad (11.4)$$

Note that, by construction, there is no mispricing at date 1 if all arbitrage capital were deployed at that date, namely, $f = 0$, because the mass of arbitrageurs is just sufficient to offset the noise trading shock. Hence, as announced earlier, the undervaluation of bond A at date $t = 1$ when a noise trading supply shock occurs is increasing in the fraction f of investors who started an arbitrage at date 0, since this will reduce the amount of speculative capital $1 - f$ left in the market at date 1.

In principle, three scenarios can arise: (1) one where $f \in (0, 1)$, i.e., some investors initiate an arbitrage at date 0 and others do so at date 1; (2) an extreme scenario where $f = 1$, i.e., all investors start an arbitrage at date 0; (3) the polar opposite case where $f = 0$, i.e., no investor starts an arbitrage at date 0.

Let us focus first on the more interesting case, namely, the intermediate scenario where $f \in (0, 1)$: this case arises when investors are indifferent between initiating an arbitrage at date 0 and doing so at date 1, i.e. $M_1 = \hat{M}_1$. Combining investors' indifference condition (11.1) with the market clearing condition (11.4) determines the fraction of investors:

$$f^* = \frac{\delta M_0}{\kappa(1 + \varphi)^2}. \quad (11.5)$$

If the expression for f^* is smaller than 1, then this is an equilibrium, in the sense that the mispricing expected to clear the market at date 1 is such that investors are indifferent between starting an arbitrage at date 0 and delaying it. This level of mispricing is:

$$M_1^* = \hat{M}_1 = \frac{M_0}{\kappa(1 + \varphi)}. \quad (11.6)$$

This equilibrium is illustrated by Figure 11.5, where the fraction f of investors who intervene at date 0 is measured on the horizontal axis and the mispricing occurring at date 1 on the vertical axis. The level of mispricing that satisfies the indifference condition, i.e. expression (11.6), corresponds to the horizontal line where $M_1^* = \hat{M}_1$. The points above this line correspond to levels of future mispricing so high that all investors would rather delay their intervention (implying $f = 0$), while the points below it correspond to levels of future mispricing so low that all of them would prefer intervening at date 0 (implying $f = 1$). Only along this line is f strictly between 0 and 1. The market-clearing condition is instead represented by the upward-sloping line in the figure: the equilibrium mispricing M_1 is increasing in the fraction f of investors who intervene at date 0, because this corresponds to a smaller fraction $1 - f$ of arbitrageurs still available to correct mispricing at date 1.

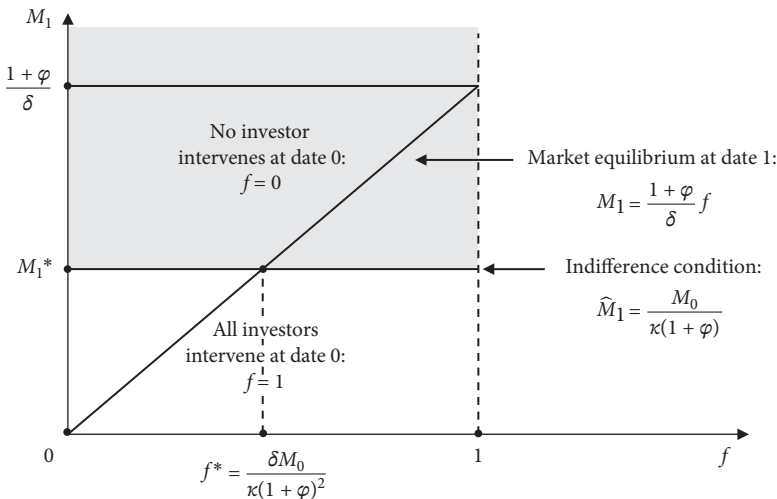


Figure 11.5 Equilibrium mispricing with some arbitrageurs intervening at date 0 and others at date 1.

Figure 11.5 helps understanding the relationship between mispricing at date 0 and at date 1. First of all, it shows that there is persistence in mispricing, as M_1^* is increasing in the initial mispricing M_0 . As M_0 increases, the indifference locus shifts upward, so that both M_1^* and f^* increase: intuitively, this is because as initial mispricing increases, it attracts scarce arbitrage capital to early intervention, causing mispricing at date 1 to increase. Hence, when arbitrage capital is limited, mispricing persists and is self-reinforcing. Interestingly, it has been observed that in derivatives markets arbitrage opportunities can be relatively persistent, taking a long time to disappear (see, for instance, Mitchell, Pedersen, and Pulvino 2007 or Gabaix, Krishnamurthy, and Vigneron 2007). This generally is ascribed to the slow entry of arbitrageurs (Duffie 2010), but it could also be due to strategic allocation of arbitrage capital over time (or more generally across various assets), as in the model developed here.

Figure 11.5 can also be used to clarify the role of noise traders and the risk of early liquidation in generating this persistence. As the probability κ of noise traders' presence and/or the risk φ of premature liquidation increase, the horizontal indifference locus shifts downward, so that f^* decreases: intuitively, more noise trading at date 1 and/or greater risk of liquidation raises the attraction of delaying arbitrage to date 1 relative to initiating it at date 0, reducing the fraction of arbitrage capital f^* deployed at date 0, and thus raising the magnitude of future mispricing M_1^* . This is also evident from expression (11.5): the fraction of investors who initiate an arbitrage at time 0 is decreasing in the risk of early liquidation φ and in the probability κ that noise traders materialize at date 1, besides being obviously increasing in the initial mispricing M_0 .

Depending on the parameters, the model can alternatively generate scenarios where in equilibrium all arbitrageurs strictly prefer to intervene either at date 0 or at date 1.

The case $f^* = 1$, where all investors decide to initiate an arbitrage at date 0, and no speculative capital is left at date 1, occurs if the initial mispricing M_0 is so large that the threshold level $\hat{M}_1$ from the indifference condition (11.1) exceeds the market-clearing mispricing given by equation (11.4) with $f = 1$, i.e., $(1 + \varphi)/\delta$. Graphically, this case occurs if the horizontal line corresponding to the indifference condition in Figure 11.5 were to lie above the upward-sloping market-clearing line for all possible values of $f \in [0, 1]$. In this case, the equilibrium mispricing would be $M_1^* = (1 + \varphi)/\delta$, obtained by setting $f = 1$ in equation (11.4), since f cannot exceed 1.

The polar opposite scenario $f^* = 0$, where no investor initiates an arbitrage at date 0, and all speculative capital is deployed at date 1, would be an equilibrium only if there is no initial mispricing, that is, $M_0 = 0$. In this case the equilibrium point would shift to the origin, where $f^* = 0$, and by equation (11.4) also the date-1 mispricing would vanish, i.e. $M_1^* = 0$. Intuitively, $M_0 = 0$ implies that nothing

can be gained from arbitrage at date 0, so all investors would delay it to date 1; but by doing so they eliminate mispricing also at date 1, because enough arbitrage capital is available to counter the negative effect of noise traders' pessimism on prices. This is because the noise traders' supply function (11.2) is such that the effect of their sales is precisely offset by arbitrageurs if all of them intervene at date 1, i.e., they were to place an aggregate buy order of size 1.

In this section we have taken date-0 mispricing M_0 as given. But this could be generated by exactly the same mechanism illustrated so far for date-1 mispricing M_1 : also at date 0 the arrival of pessimistic investors can push the price below fundamental value. The exact level of date-0 mispricing will then depend on how much arbitrage capital is allocated to intervention at date 0 (we leave this analysis as exercise 2). In general, mispricing at date 0 will not disappear because in our model the arbitrage capital available at this date is limited: the prospect that the mispricing of the bond may be greater at date 1 induces some arbitrageurs to wait rather than exploit the opportunity at date 0.

In reality, arbitraging activity may be curtailed not only by the risk of forced liquidation, but by three other frictions as well.

First, arbitrageurs may be risk averse, unlike in our analysis. The cash flows generated by arbitrage portfolios are uncertain when there is a risk of early liquidation, as in our model, and this risk may not be easily diversified if arbitrageurs are specialized (say, because their activity requires expertise).

Second, as noted above, arbitrageurs do not earn the market interest rate on the collateral that they are required to post on their short positions. This point is not captured by our model, which posits a zero market interest rate. When these costs are large relative to the benefit of an arbitrage opportunity, arbitrageurs will simply refrain from taking advantage of it, and the opportunity will persist.

Thirdly, in the model presented above, the number of arbitrageurs that choose to be active either at date 0 or at date 1 is fixed. In reality, the presence of an arbitrage opportunity may attract the attention of outside investors whose intervention will help to close the arbitrage opportunity. However, in some cases it may take time for these further investors to become aware of the mispricing and intervene, so that the mispricing will persist for a long time: Duffie (2010) characterizes this situation as one of "slow-moving capital" and explains that it can reflect institutional impediments, such as search costs for trading counterparties or time to raise capital by intermediaries. Accordingly, Duffie, Gârleanu, and Pedersen (2007) show that the speed with which transaction prices recover after "fire sales" depends to a large extent on search costs and the liquidity of the market: in the case of illiquid environments, price recovery may take considerable time as market participants have to wait for the arrival of counterparties. That is, limits to arbitrage are themselves a function of market liquidity.

11.2.3 Implications for Market Making

As highlighted in the introduction to this chapter, the arbitrageurs modelled in the previous sections can be viewed as providing liquidity to noise traders. For instance, arbitrageurs with capital at date 1 are “leaning against the wind” by absorbing the sell orders for asset A of noise traders and the fire sales of arbitrageurs who liquidate at that date. In doing this and selling asset B short, they act exactly as market makers, who fill sell orders and simultaneously hedge their positions.¹⁰ Thus, one can interpret the magnitude of the mispricing M_1 as the price impact of sell orders due to the lack of liquidity. Indeed, if all arbitrage capital is deployed at date 1, the market at that date is fully liquid, in the sense that it absorbs noise traders’ sell orders with no discount.

This interpretation highlights that arbitrageurs’ activity closely parallels market-making activity. Market participants such as hedge funds, which take and hold positions for long periods, play a similar role conceptually to that of the market makers described earlier (e.g., NYSE specialists or Nasdaq dealers). The main difference is that they do not operate at the same frequency. They take long positions in underperforming stocks and short ones in overperforming stocks (see Khandani and Lo 2011), seeking to exploit the respective price reversals towards fundamentals. Like the market makers analyzed in Chapter 3, they take inventory risks stemming from uncertainty regarding the extent and the speed of price reversals, the main difference with market makers being that they seek to exploit price reversal at a lower frequency. The analogy between the role of arbitrageurs and market makers is highlighted by Grossman and Miller (1988) and Huang and Wang (2009, 2010).¹¹

11.3 Correlated Order Flow and Noise Trader Risk

In the previous section, noise trading was seen to be an important trigger of mispricing, because the price pressure generated may be so great that arbitrageurs are unable to absorb their orders and keep market prices from diverging further from fundamentals. One reason is that noise traders’ orders tend to be highly

¹⁰ For instance, a market maker taking a long position in a stock can hedge the systematic risk by selling futures on a stock index.

¹¹ They present a model of market making with inventory risk. If a large number of investors decide to sell a security for liquidity reasons, its price will drop to attract buyers even though its cash flow is unchanged. This temporarily low price makes it attractive for investors to step in and buy, providing liquidity to the sellers—effectively playing the role of market makers. If there are many such investors, large sell orders have little or no impact on price. But if these liquidity suppliers have limited capital, their ability to absorb orders may be impaired, in which case the price will be sensitive to sell orders.

correlated, so that they sell or buy in waves. Dorn, Huberman, and Sengmueller (2008), using data on a sample of retail investors with accounts at a large online broker in Germany, find that their trades are positively correlated.

This can either reflect correlation of their liquidity needs or a tendency to follow the same trading rules (reacting in the same way to the same information). Liquidity needs may be correlated because of macroeconomic shocks to many households at once. During the COVID-19 pandemic, for instance, social distancing rules have reduced many households' spending on leisure activities, such as travel, restaurant meals, and live entertainment; this has led them to accumulate abnormal savings, which they have invested in assets, such as securities and real estate, inflating their prices. Moreover, investors may display "herd" behavior, in the sense that they may imitate one another's trading strategies, stampeding into or out of specific investments. Nowadays, noise traders can also use social media to coordinate their trades in real time, as exemplified by the GameStop saga (see Box 11.2). Alternatively, they may follow common mechanical trading "momentum strategies," which lead them to sell at the same time when securities prices are falling (and buy when they are rising). This type of behavior is modelled by De Long et al. (1990) by assuming that noise traders behave as "positive feedback traders," who buy when prices rise and sell when they fall.

A simple way to grasp this point is to consider the inventory model of Chapter 3, where the dealers' quotes are centered on the price $m_t = \mu_t - \rho\sigma^2 z_t$, which decreases with the dealers' risk aversion ρ , the riskiness of the security σ^2 , and the dealers' inventory z_t . In that setting, larger orders (large $|q_t|$) induce greater imbalance in dealers' inventory ($|z_t|$), and thereby in their risk. If the orders of noise traders are correlated, the swings in their trade sizes, and hence in z_t and m_t , are larger. Seen ex ante through the eyes of investors, this phenomenon tends to increase risk: each of them should consider a stock to be riskier if many others can be expected to sell when he wants to sell. Hence, noise traders' behavior is in itself a source of risk, distinct from the fundamental risk due to the arrival of new information regarding the asset payoff (i.e., changes in μ_t).¹²

Interestingly, rational speculators—or arbitrageurs—may not offset the price swings induced by feedback noise traders or by dynamic hedgers, but rather exacerbate them. In De Long et al. (1990), this is because speculators may gain by jumping on the bandwagon and trading ahead of the noise traders; in Gennotte and Leland (1990) it is because they may incorrectly interpret selling by dynamic

¹² Building on this observation, several studies attempt to quantify the contribution of noise traders to the volatility of stock returns (see, for instance, Foucault, Thesmar, and Sraer (2011) or Hendershott et al. (2010)).

hedgers as driven by bad news, inducing them to join the latter in a wave of selling.

This risk induced by noise trading is greater in thin markets, which cannot absorb large bulges of buy or sell orders without wide price fluctuations. Pagano (1989a) and Allen and Gale (1994) show that in certain circumstances noise trader risk can trap the market into a self-perpetuating high-volatility, low-volume equilibrium, where noise trader risk deters new investors and low volume makes for acute sensitivity to noise traders' orders.

Box 11.2 The GameStop Short Squeeze of January 2021

The GameStop episode of early 2021 illustrates the dangers of risky arbitrage, namely the prospect of being forced to close the position prematurely before the stock price reaches its predicted long-run level.¹³ In 2020 a number of hedge funds, pessimistic about GameStop's prospects, heavily shorted the stock: the short interest reported over the third quarter of 2020 ranged from 66.8 to 70.3 million shares, representing over 100% of the 65.2 million shares issued. Thus conditions were ripe for a classic short squeeze, in which the supply of stock is cornered, forcing short sellers who are obliged to deliver stock to buy it in at an inflated price.¹⁴ Even if the dire forecasts for GameStop's business prospects were to come true eventually, that would be small comfort to short sellers who were forced to liquidate their position before the long awaited stock price correction materializes.

In early January 2021, a series of online posts on Reddit whipped gaming aficionados into action to teach the short sellers a lesson and protect GameStop, a retailer of the gaming merchandise that they valued. As shown by Figure 11.6, the buying frenzy that ensued in the second half of January drove up the GameStop stock price, calibrated on the left-hand axis, to stratospheric heights. It briefly reached \$483 on January 28, for a stock that had been trading in the single digits for most of 2020, and many of the short sellers were forced to close out their positions at a crippling loss, and some to exit permanently from short selling altogether. Short interest in GameStop plunged from 62 to 21 million shares in the second half of January, and continued to decline steadily after that, to about 10 million shares in mid-March,

¹³ GameStop Corp. is a large American bricks-and-mortar video game, consumer electronics, and gaming merchandise retailer.

¹⁴ The SEC's Rule 204 deals with settlement failures by requiring a forced buy-in of stock at any price so as to ensure that it is delivered on time. The rule was instituted in 2005 after a long history of abuses: failures to settle were common, and such failures are tantamount to forcing the short-seller's counterparty to extend a free loan of stock.

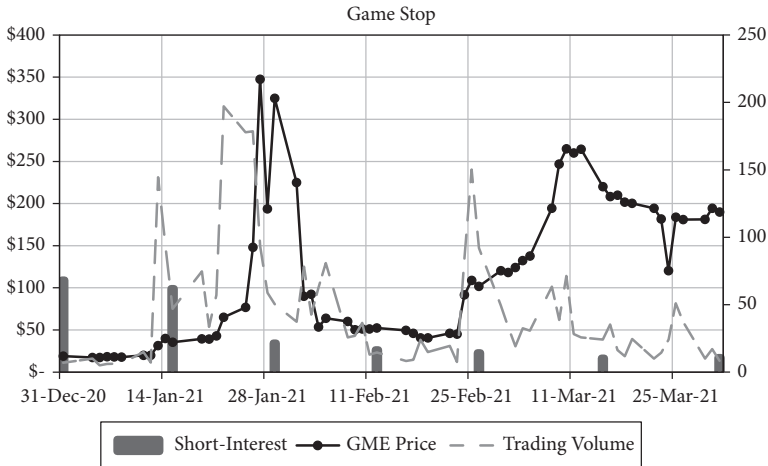


Figure 11.6 GameStop stock price, trading volume and short interest in the first quarter of 2021.

as depicted in Figure 11.6, where short interest and trading volume are measured in millions of shares of the right-hand axis.

One of the novel features of this episode was that the short squeeze involved concerted action by small, unsophisticated traders, fueled by two recent developments. First, the rise of social media such as Reddit enabled direct real-time commentary and coordination, giving rise to ringside excitement far beyond that provided by traditional, less immediate, methods of generating interest in stocks (unsolicited phone calls, newsletters, faxes). Second, commission-free, user-friendly trading apps offered by Robinhood and other brokers made stock trading easy, fast, and ostensibly costless, drawing inexperienced investors into excessive leveraged speculation. These developments have fueled a rash of “meme” stocks caught up in sudden bursts of retail investor excitement, of which GameStop is but one example.

Many of those who bought at the height of the price spike around the end of January lost a large proportion of the money they invested at the behest of their supposed peers—some of whom profited handsomely and may now face class action lawsuits for misrepresenting themselves as amateur investors. And investors were deeply suspicious of the motives of Robinhood in its moves to calm the excesses by raising margins and restricting purchases of GameStop stock. At the height of the frenzy, the GameStop closing price rose by 135% in a single day, Wednesday, January 27. Robinhood imposed trading restrictions on January 28 which contributed to the 44% fall in the

closing price between January 27 and 28: by the close, the price had fallen to \$193 from its intraday high of \$483. Robinhood justified its actions citing dramatic increases in collateral required by its clearinghouse: it was forced to raise money to meet these, drawing down over \$500 million in lines of credit, and raising \$1 billion from its existing investors late on Thursday, January 28. Several other brokers followed suit. After the Thursday market close, Robinhood announced the restrictions would be loosened, precipitating a 68% price spike on that Friday. A subsequent investor lawsuit against Robinhood accusing it of colluding with Citadel Securities, which paid it to execute orders, to stop investors from buying on January 28 was dismissed by a federal judge in November 2021.

Concerns about market manipulation and the protection of unsophisticated investors from “pump and dump” schemes led to a series of congressional hearings starting on February 18, 2021, which, interestingly, precipitated a renewed wave of buying pressure triggered by the defiant testimony of the protagonists. The relevant investor protection issues are wide ranging, including the use of social media in encouraging small investors to buy, and the role of Robinhood in fueling the price gyrations: its profits from payment-for-order-flow by market makers—in particular, Citadel Securities—are driven by volatility and trading volume.

By late March analyst forecasts for GameStop were up, a boardroom shakeup was announced (the departure of the failed CEO would entail vesting over 1.1 million shares, giving him an outlandish exit payout of over \$200 million at reigning stock prices) together with plans to raise equity capital (by the end of June 2021 the company was to sell 8.5 million shares, raising over \$1.6 billion) and improve the company’s business model. Though prices failed to revisit the stratospheric levels reached at the end of January they remained far above the levels reached before the short squeeze. The cause of this enduring price rise is an open question: have potential short sellers been chastened enough to steer clear of the stock despite their negative views of its fundamental value, or has there been a marked improvement in the company’s business model and long-term prospects?

The GameStop frenzy is a perfect example of correlated noise traders’ order flow and of its ability to overwhelm arbitrageurs’ trading strategies. It illustrates that noise traders may determine not only unwarranted stock market crashes but also booms, which cause short squeezes, i.e., inflict losses on arbitrageurs who short sell when they think a stock is overvalued. It also illustrates that such correlated behavior has been enhanced by noise traders’ ability to rely on social media to communicate and coordinate their orders.

11.4 Feedback Loop between Market and Funding Liquidity

A key insight from the limits-to-arbitrage model presented above is that arbitrage requires external funding: a shortage of external funding can force arbitrageurs to scale down their activity or even withdraw completely from some markets, leading to wider mispricing and illiquidity. This explains why a credit crunch affecting market makers often leads to reduced liquidity in security markets: in the words of Brunnermeier and Pedersen (2009), a sudden drying up of “funding liquidity” reduces market liquidity as well.¹⁵

Market makers can face a drying up of funding liquidity just when it is most needed, precipitating a very sharp fall in asset prices. In turn, a sharp drop in assets prices may jeopardize the ability of banks to provide funding to market makers: for instance, it may reduce the value of banks’ assets to the point where they hit a regulatory capital constraint or it may stoke doubts about the solvency of banks among fixed-income investors or depositors. This may generate a feedback loop between market liquidity and funding liquidity, as shown by Figure 11.1: an increase in mispricing triggers a drop in funding by banks, which in turn exacerbates the mispricing in the market. Brunnermeier and Pedersen (2009) refer to this feedback loop as an “illiquidity spiral.”

To capture this point in the model of the previous section, suppose that if mispricing at time 1 rises above a critical threshold $\bar{M}_1$, then banks curtail funding to a fraction φ' of arbitrageurs larger than the fraction φ initially anticipated. This can be because the drop in asset prices hits their own balance sheet (possibly to the point of triggering their bankruptcy) and therefore curtails their lending capacity. In other words, from the standpoint of an arbitrageur that opened a position at time 0, there is an unanticipated increase of the risk of facing premature liquidation from φ to φ' . A good example of such an unanticipated collapse in credit is that triggered by the Lehman bankruptcy in September 2008: as Lehman collapsed unexpectedly, it withdrew credit to hedge funds that it was previously funding, and this shock reduced the market liquidity of the assets that those funds were trading (Aragon and Strahan 2012).

We will show that this unexpected drop in funding liquidity may lead mispricing to become so large as to exceed the banks’ critical threshold, i.e. to $M_1 > \bar{M}_1$, and thus justify their decision to curtail funding to arbitrageurs more than initially anticipated. In other words, banks’ belief of extreme mispricing can be self-fulfilling, i.e. generate a drop in funding to arbitrageurs that determines

¹⁵ Here, a market crash is triggered by a reduction of the funding liquidity by banks and of market liquidity by market makers. But in principle, it can also arise from a scramble for liquidity by investors: Huang and Wang (2009) present a theory of market crashes based on the idea that the demand for liquidity may become suddenly too great relative to the supply.

precisely such an extreme mispricing. We refer to this situation as a “crisis equilibrium,” and will see that it can coexist with an equilibrium featuring a “normal” level of mispricing such as that shown in Figure 11.5.

We assume that crises are rare events: as of time 0, the probability of a crisis occurring at time 1 is very low. Hence arbitrageurs treat a crisis as a zero-probability event in the choice of their strategy at date 0, neglecting the possibility that φ may increase to φ' at date 1. Hence, f^* is determined by equation (11.5), as before.¹⁶

If the economy is not in a crisis, the equilibrium mispricing M_1^* is determined by equation (11.6), as in Figure 11.5. If a crisis does occur however, the clearing condition (11.4) is altered since a fraction $\varphi' > \varphi$ of the arbitrageurs who did not intervene at date 0 are forced to liquidate their positions prematurely. Hence, the equilibrium mispricing in a crisis is:

$$M_1^{**} = \frac{1 + \varphi'}{\delta} f^* = \frac{1 + \varphi'}{1 + \varphi} M_1^* > M_1^*. \quad (11.7)$$

Note that, for this to be a “crisis equilibrium,” the mispricing in equation (11.7) triggered by the drop in funding must be so large as to jeopardize banks’ funding ability, namely, $M_1^{**} > \bar{M}_1$. If this condition were violated, banks would never react to the increase in mispricing by further curtailing funding, so that a crisis could not occur. Compared to its value in the “normal-times equilibrium,” given by equation (11.6), asset A is more severely mispriced in the “crisis equilibrium,” the percentage increase in mispricing being approximately proportional to the incremental drop in funding: $(M_1^{**} - M_1^*)/M_1^* = (\varphi' - \varphi)/(1 + \varphi)$: the sharper the dry-up in bank funding, the more severe the drop in the market price at date 1.

The two equilibria are illustrated in Figure 11.7, where f^* represents the allocation of arbitrage capital between the two dates, which does not change, as it was determined giving a negligible weight to the possibility of crisis; the level of mispricing in a crisis, M_1^{**} , is found on the new market-clearing locus (11.7) in correspondence with the original value of f^* . As Figure 11.7 indicates, the increase in mispricing triggered by the crisis—the market crash—will be more dramatic, the larger the drop in funding liquidity $\varphi' - \varphi$.

The coexistence of a normal-time equilibrium and a crisis equilibrium arises from the feedback loop between market liquidity and funding liquidity present in this model illustrated by Figure 11.1 in the introduction to this chapter: the initial noise trading supply shock moves prices below fundamentals at time 1; if

¹⁶ This assumption is made for analytical simplicity: the results obtained in this section would be qualitatively unchanged if arbitrageurs were to consider a crisis as an event with positive probability.

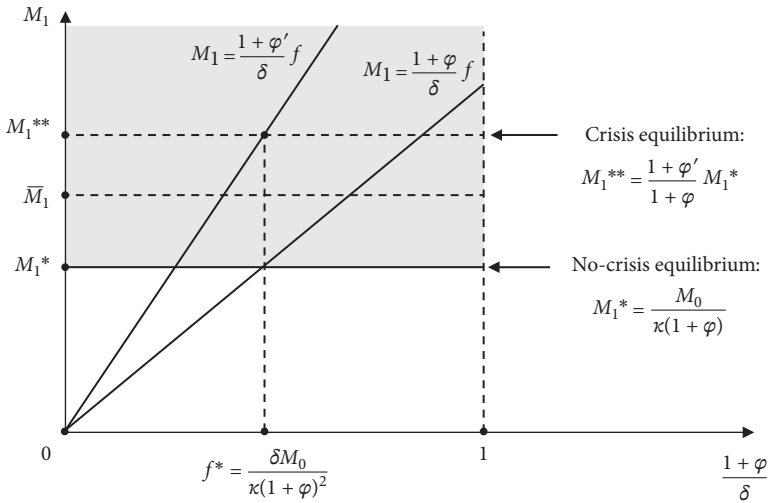


Figure 11.7 Mispricing and allocation of arbitrage capital with and without crisis.

banks are exposed to the asset, they suffer capital losses as a result of the drop in price, and this precipitates a drop in credit to arbitrageurs, who will consequently scale back their trading. This will in turn exacerbate the mispricing, leading to another round of bank losses, credit crunch, and arbitrageurs' withdrawal from the market. This vicious circle lies at the heart of the crisis equilibrium.

This type of feedback loop is modeled more exhaustively by Brunnermeier and Pedersen (2009), who introduce a second feedback channel: insofar as mispricing generates greater expected volatility, it triggers higher margin requirements for speculators, forcing them to reduce their positions, which in turn generates further mispricing. A similar mechanism can arise even in the absence of margin requirements, if speculators are risk averse: high volatility can then trigger a rise in bid-ask spreads, which in turn adds to the volatility of transaction prices (as explained in Chapters 3 and 4), generating a feedback loop akin to that modeled by Brunnermeier and Pedersen (2014).

An important issue is why financial intermediaries respond to increased asset mispricing by curtailing credit to arbitrageurs, and hence precipitate the extreme increase in mispricing that is typical of a financial crisis. In the above notation, why does the expectation of $M_1^{**} > \bar{M}_1$ lead banks to raise the frequency of liquidation from φ to φ' ?

One answer is that as mispricing rises, banks will need to mark down their loans to arbitrageurs, in line with the drop in the value of their collateral; moreover, banks may suffer capital losses on their trading book, insofar as their security portfolios are themselves exposed to the mispriced asset. On both

accounts, banks will suffer capital losses on the asset side of their balance sheet, and in order to comply with regulatory capital ratios they will have to reduce lending. Prudential regulation imposes such capital requirements on banks precisely on account of their intrinsic fragility, stemming from the maturity transformation that they perform: banks use demand deposits to fund longer term loans, and therefore are themselves exposed to “runs” by depositors, as shown by Diamond and Dybvig (1983). When depositors fear that the payoff of the loans funded by their bank may be insufficient to convert all of their deposits into cash, they have the incentive to withdraw their deposits before the others do. This will in turn force the bank to recall its loans and liquidate its securities at fire-sale prices, thus validating depositors’ pessimistic conjecture. In other words, banks themselves are fragile, as their maturity transformation function creates strategic complementarity between depositors, which in turn generates equilibria with runs (beside those with no runs).

Banks are not the only intermediaries exposed to runs: mutual funds are also exposed to runs by their shareholders. In general, insofar as mutual funds invest in diversified bundles of assets featuring different liquidity, they will tend to sell first their most liquid assets (those with the lowest bid-ask spreads and price impact) in response to withdrawals by their shareholders. As a result, when shareholders expect abnormal withdrawals by others, they have the incentive to be first in line in withdrawing their own funds, to avoid being left holding a bundle of less liquid shares, which are going to sell for less upon liquidation. As in the case of banks, these runs arise from strategic complementarities among investors. Consistent with this idea, Chen, Goldstein, and Jiang (2010) find that mutual funds with illiquid assets (where complementarities are stronger) exhibit stronger sensitivity of outflows to bad past performance than funds with liquid assets. Relatedly, Ben-Rephael (2017) documents that at times of extreme market uncertainty, mutual funds reduce their aggregate holdings of illiquid stocks, as a result of larger withdrawals from funds holding such stocks.

11.5 Contagion, Flight to Safety, and Systemic Instability

The mispricing modeled in the previous section with reference to a single asset (bond *A* in our example) often involves multiple assets at the same time: mispricing can be local, i.e., involve a specific asset, or systemic, i.e. spread across asset classes or intermediaries, which is the key feature of a financial crises. This systemic feature of mispricing may arise either from “commonality” in the liquidity shocks, i.e., their simultaneous effect on many different markets and intermediaries, or from their tendency to propagate across financial intermediaries via chains of lending relationships, or from both mechanisms.

The occurrence of a common liquidity shock can explain why, in the model of the previous section, banks may wish to terminate loans to arbitrageurs precisely at the time when noise traders materialize and generate selling pressure on the security market. The reason may be that banks themselves are hit by a concomitant liquidity shock, such as a run by their depositors. Even if arbitrageurs are not funded by bank loans, but by short-term debt securities such as commercial paper, they may be unable to roll over such funding when they face selling pressure, because the commercial paper market may also dry up (or feature a spike in yields) at the same time due to sales by noise traders occurring also on that market.

Even a liquidity shock that hits a single market can propagate across markets and intermediaries, and generate global, rather than local, mispricing. A clear example of this occurred in the 2008 global financial crisis, which originated in the subprime mortgage lending market, a very specific corner of the credit market, but almost metastasized into a complete financial meltdown. The main reason for such propagation is that financial intermediaries (mutual funds, hedge funds, and market makers, as well as the banks who fund them) are often highly interconnected. For instance, suppose a group of arbitrageurs face premature withdrawals of funding and are forced to liquidate their positions. The drop in security prices that they trigger can inflict losses on other intermediaries: for example, if a bond is used by banks as collateral for their loans, a large drop in its price can make the collateral insufficient to guarantee existing loans and prompt banks to recall them, forcing another round of premature asset liquidations by their borrowers. Hence, withdrawals of funds and fire sales can propagate both across asset classes and across different groups of market participants. In the most extreme instance, this domino effect may precipitate an economy-wide financial collapse, which ultimately requires intervention of the central bank acting as “lender of last resort” to distressed intermediaries in the form of an extraordinary credit expansion and/or massive open market bond purchases.

A specific example of such contagion among financial institutions is the effect of the collapse of the Lehman Brothers investment bank on U.S. money market funds in 2008. Since these funds invest in very safe short-term debt, such as T-Bills, their per-share net asset value (NAV) normally stays constant at \$1: this is facilitated by market regulation, which allows these funds to value their assets at cost rather than market value. Hence, U.S. investors treat their shares of money market funds as equivalent to bank deposits. But, following the bankruptcy of Lehman Brothers in September 2008, the NAV of the prominent Reserve Primary Fund fell to 0.97 cents per share (it “broke the buck”) because it had invested 1.5% of its assets in Lehman commercial paper, and worried investors started redeeming their holdings, so that the fund’s assets declined by nearly

two-thirds in 24 hours. Being unable to meet redemption requests, the fund was forced to liquidate. More generally, investors began to doubt the safety of money market funds as a safe, liquid place to invest, and the panic extended to other money market funds that had invested in the commercial paper of banks. This contributed to the dry-up of liquidity in the commercial paper market, which propagated the crisis to yet more financial institutions.

Network analysis is often used to model the interconnections between financial institutions, and thus understand how resilient financial markets are to such forms of contagion. The connections between financial institutions stem from both the asset and the liability sides of their balance sheets, and their connections can be either direct or indirect. For instance, banks are directly connected via mutual exposures on the interbank market; at the same time, they may be indirectly linked if they hold similar security and/or loan portfolios or are funded by the same depositors.

The structure of the network can affect its resilience to financial shocks. Allen and Gale (2000) show that a more densely interconnected network enhances the resilience of the system to the insolvency of any individual bank, as the losses of a distressed bank are divided among more creditors, reducing the impact of its insolvency on the rest of the system, compared to a network where a few institutions play a central role. This underscores the importance of identifying these “systemically important financial institutions” (SIFI), i.e., those whose failure might precipitate a financial crisis, so as to ensure their solvency with higher capital requirements than other banks. Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015) confirm the insight that a more diversified pattern of interbank liabilities leads to a less fragile financial system, provided the magnitude of negative shocks is below a certain threshold, but also show that, faced with sufficiently large shocks, highly diversified financial networks facilitate financial contagion and thus lower systemic stability. The intuition is that, when negative shocks are very large, the excess liquidity of the banking system may no longer be sufficient to absorb the resulting losses. In this case, financial interconnections act as a propagation mechanism, rather than as a stabilizing one. In contrast, a less diversified lending pattern guarantees that the losses are shared with the senior creditors of the distressed banks, rather than transmitting the shocks to other banks in the system.

Contagion helps to account not only for the propagation of mispricing, fund withdrawals, and insolvencies but also for the “commonality” in market liquidity, namely, the tendency of bid-ask spreads to widen simultaneously for many securities, as funding to market makers is curtailed across the board. The commonality in bid-ask spreads is reinforced by the commonality of risk: at times of market turbulence, the risk of assets with correlated returns rises

simultaneously, leading risk-averse market-makers to widen their spreads simultaneously for all of them, risk being a key driver of bid-ask spreads, as seen in Chapter 3.

11.6 Adverse Selection and Financial Instability

The feedback loop between funding and market liquidity modeled by Brunnermeier and Pedersen (2008) is not the only explanation for financial crises proposed in the literature. In particular, financial crises have been viewed as situations in which adverse selection problems become so extreme as to generate very high illiquidity or even market collapse.

Adverse selection regarding the quality of an asset may suddenly increase if confidence in the issuer of the asset declines and people start researching the asset's fundamentals. If only a few investors manage to get hold of the information and interpret it correctly, adverse selection will intensify: the asset will become more information-sensitive, and as a result its market will become less liquid. Insofar as this induces unsophisticated traders to withdraw from the market, it may lead to market collapse (Pagano and Volpin 2012 and Dang, Gorton, and Holmstrom 2020). This has been proposed as one possible explanation for the collapse of the U.S. market for securitized mortgage debt in 2008.

Market collapse due to increased adverse selection may also arise from counterparty risk, namely, the probability that the other party in a transaction may default on its contractual obligations. While so far in this book adverse selection has been assumed to concern only the payoffs of assets, in practice it may also concern the solvency of the intermediaries an investor is trading with, hence the counterparty risk the investor is exposed to. When market turbulence increases, the probability that a given financial intermediary may be unable to meet its obligations increases too. This may result in investors losing confidence in the solvency of their potential counterparties and withdrawing from trading. In the extreme, it may lead to market collapse. Heider, Hoerova, and Holthausen (2015) propose a model where an increase in counterparty risk can precipitate a crisis, and use it to interpret developments in interbank markets prior to and during the 2007–9 financial crisis.

Counterparty risk is typically mitigated by central clearing counterparties (CCPs), namely, institutions that take on counterparty credit risk between parties to a transaction and provide clearing and settlement services for trades (see Chapter 7). CCPs share counterparty credit risk among their members in the markets in which they operate. They reduce settlement risks not only by

netting offsetting transactions between multiple counterparties,¹⁷ but also by requiring collateral deposits (called “margin deposits”) and often by creating a guarantee fund to be used to cover losses in excess of a defaulting member’s collateral on deposit.

Multilateral netting reduces the cost of posting collateral, as a given amount of collateral can support a greater number of positions: Duffie and Zhu (2011) argue that this is the key comparative advantage of CCPs over a system where counterparties agree bilaterally on the collateral to be posted as protection against counterparty risk (provided there is “interoperability” across CCPs, that is, links that allow for multilateral netting across them). Hence, CCPs are a cost-efficient way of mitigating adverse selection (as well as moral hazard) issues arising from counterparty risk.¹⁸ A further advantage of CCPs lies in their ability to mitigate fire sales, insofar as they liquidate the securities of members who default on their margin calls via auctions among their surviving members, rather than in the open market. These auctions can be seen as a mechanism to reduce the inefficiencies associated with fire sales due to the scarcity of speculative capital analyzed earlier in this chapter.

The benefits of CCPs are illustrated by Vuillemeys (2020) with reference to the creation of the first CCP in Le Havre (France) in 1882, in the coffee futures market. He shows that central clearing changed the geography of trade flows in Europe, to the benefit of Le Havre, and as a result CCPs quickly spread to other exchanges. Of course, the benefits of CCPs are conditional on the CCPs themselves not collapsing. When a large enough fraction of transactions is centrally cleared, as it is now generally the case, the solvency of CCPs becomes crucial to the smooth functioning of security markets. The risk of a CCP failure is not a remote possibility: during the recent COVID-19 crisis, in as many as half a dozen CCPs the clearing members’ mark-to-market losses exceeded the collateral they posted by more than 1%, which is the industry standard (Menkveld and Vuillemeys 2020).

11.7 Further Reading

The theoretical literature on limits to arbitrage can be traced back to Dow and Gorton (1994) and Shleifer and Vishny (1997). Kondor (2009) develops a continuous-time model of convergence trading in which arbitrageurs do not

¹⁷ Netting enables an investor to offset losses in a position with gains in another. For example, an investor who is short 40 shares of a security and long 100 shares of the same security vis-à-vis different counterparties has a net long position of 60 shares.

¹⁸ See Kuong and Maurin (2022) for a theory of the optimal design of CCPs.

immediately allocate all their capital to an opportunity, anticipating that it may become even more profitable in the future. He shows that, in equilibrium, the size of the opportunity must necessarily increase with positive probability over time. Gromb and Vayanos (2002) explicitly model the link between arbitrageurs' past performance and the amount of arbitrage capital available at a given date, in the presence of margin requirements. If performance is bad, arbitrageurs' wealth declines. Thus, margin requirements limit the amount of capital that can be devoted to arbitrage activities more stringently, further depressing prices. Brunnermeier and Pedersen (2008) extend their model to the case of multiple arbitrage opportunities, showing how financial constraints (such as margin requirements) can generate co-movement in liquidity in various securities. Duffie (2010) surveys recent research on these price effects across multiple markets associated with slow-moving capital. Duffie and Strulovici (2009) model the equilibrium movement of capital between asset markets with different levels of invested capital, and show that the greater the difference in capital levels, the greater the intermediaries' effort to re-balance the distribution of capital across markets, and the faster the convergence of the mean rates of return on different assets toward a common level. Xiong (2001) and Kyle and Xiong (2001) consider a dynamic setting in which arbitrageurs have logarithmic utility functions, so that their demand for risky securities increases with their wealth. As a consequence, losses on arbitrageurs' positions induce a decrease in their demand for risky securities and further price decline. Gromb and Vayanos (2010) survey the theoretical literature on the limits to arbitrage.

Many of the empirical studies on limits to arbitrage analyze the temporary underpricing and subsequent price reversal induced by "fire sales." Coval and Stafford (2007) investigate stock market transactions induced by the redemptions of open-end mutual funds, and find that funds undergoing large outflows tend to decrease the size of their positions, which puts pressure on prices of the securities held in common by distressed funds. Mitchell, Pedersen, and Pulvino (2007) study the price reactions of convertible bonds to forced redemptions of hedge funds. Pulvino (1998) shows that financially constrained airlines get lower prices than their unconstrained rivals when they sell used narrow-body aircraft. Campbell, Giglio, and Pathak (2011) show that forced sales of real estate due to foreclosures are associated with substantial price discounts followed by mean-reversion, while unforced sales prices are close to a random walk. Ellul, Jotikasthira, and Lundblad (2011) examine fire sales of speculative-grade corporate bonds by insurance companies arising from regulatory constraints and/or capital requirements. They find that more severely constrained companies are, on average, more likely to sell downgraded bonds, and that bonds whose probability of regulatory-induced selling is higher exhibit significant price declines and subsequent reversals, especially when insurance

companies as a group are more distressed and other potential buyers have scarce capital.

A vast literature applies network theory to the analysis of financial contagion and systemic stability. Beside the theoretical studies by Allen and Gale (2000) and Acemoglu et al. (2015) cited in the text, a growing empirical literature applies network-theoretic approaches to the analysis of systemic stability of interbank and securities markets: for instance, Furfine (2003) uses a network-based approach to analyze solvency contagion in the U.S. federal funds market, and Drehmann and Tarashev (2013) use it to measure the systemic importance of interconnected banks.

The commonality in liquidity is well-documented both across stocks (Chordia, Roll, and Subrahmanyam 2000; Hasbrouck and Seppi 2001; and Huberman and Halka 2001), and across stocks and bonds (Chordia, Sarkar, and Subrahmanyam 2005). There is also evidence that connects commonality in liquidity to the availability of credit to market makers (Acharya, Schaefer, and Zhang 2008). Coughenour and Saad (2004) find that the comovements in liquidity for stocks handled by the same specialist on the NYSE are explained both by movements in the liquidity of the relevant specialist's portfolio and by market-wide movements in liquidity. Hence, increases in the cost of capital or increased risk exposure for a specialist leads to a drop in liquidity for the stocks assigned to that specialist.

While this chapter highlights that the mutual feedback between funding and market liquidity may generate financial instability, further mutual feedbacks can develop between financial instability and the real economy. For instance, a financial crisis typically triggers a recession, which results in firm bankruptcies, hence losses by banks and other financial intermediaries, and further retrenchment of credit. These further feedback loops and the resulting non-linearities in the response of the economy to shocks have been the object of a vast macro-finance literature, such as He and Krishnamurthy (2012, 2013), Brunnermeier and Sannikov (2014), and Brunnermeier et al. (2016, 2017). The concern that financial crises may result in real instability and vice versa has led to much recent research regarding macro-prudential policies, i.e., financial policies aimed at reducing the probability of financial crises and/or mitigate their consequences: macroprudential capital buffers for banks, ceilings to loan-to-value (LTV) ratios for real estate lending, minimum liquidity ratios and redemption gates for mutual funds. A key issue in the design of macro-prudential policy is to avoid the pro-cyclicality of micro-prudential policy rules, for instance preventing the capital ratios imposed on banks to ensure their solvency from inducing them to liquidate assets and call back loans in the middle of a severe recession. Freixas, Laeven, and Peydró (2015) provide an early treatment of these topics, while Duffie (2018) examines how capital and failure-resolution rules for systemically

important banks adopted after the 2008–9 financial crisis have affected the liquidity of security markets, especially OTC ones.

11.8 Exercises

1. **Maximal mispricing.** Suppose that, in the model of limits to arbitrage presented in Section 11.2.2, the parameters are such that $\delta M_0 > \kappa(1 + \varphi)^2$. What is the resulting equilibrium allocation of arbitrageurs across periods, and the implied equilibrium mispricing at date 1? Provide an intuitive explanation.
2. **Endogenous mispricing at dates 0 and 1.** Consider the model of arbitrage in Section 11.2.2. Assume that, at date 0, the total supply from noise traders is:

$$y(P_{A0}) = 1 + \delta_0(P_{A0} - V), \quad (11.8)$$

where P_{A0} is the price of asset A at date 0. Otherwise the model is unchanged.

- a. Write down the system of equations that determines M_0 , M_1 , and f in equilibrium.
- b. Compute the equilibrium values of mispricing at dates 0 and 1, M_0^* and M_1^* , and the equilibrium fraction of arbitrageurs intervening at date 0, f^* .
- c. Consider how the parameters κ , δ , δ_0 , and φ affect the equilibrium mispricing at each of the two dates and the proportion of arbitrageurs intervening at the two dates, and provide an intuitive explanation for your findings.
3. **Carry-trade arbitrage.** In this exercise we analyze the returns and risks of a currency carry trade. A carry trade involves borrowing money in a low interest-rate currency and investing it in a high interest-rate currency. Based on the theories presented on liquidity risk and limits to arbitrage, we will re-evaluate the attractiveness of investing in such a strategy. The Excel file Ch11_ex5_data.xlsx, available on the companion website for the book, contains data on the monthly risk-free rate in Australian Dollars (AUD) and Swiss Franc (CHF) and the CHFAUD exchange rate (number of Swiss Francs per Australian Dollar), denoted x_{CHFAUD} .¹⁹ It also contains data on the S&P 500 monthly return, the average U.S. stock bid-ask spread and the TED-spread. These variables will be used in the later part of the exercise.

¹⁹ Source: Bloomberg.

- a. A popular carry trade is to borrow CHF at the risk-free rate, exchange them to AUD and invest at the AUD risk-free rate. The monthly return (in Swiss Francs) from this strategy can be approximated as:

$$R_t \approx r_{AUD,t-1} - r_{CHF,t-1} - \frac{\Delta x_{CHFAUD,t}}{x_{CHFAUD,t-1}},$$

where $r_{y,t}$ indicates the risk-free rate in currency y in month t and $\Delta x_{CHFAUD,t}$ the change in exchange rate from the end of month $t-1$ to month t . Calculate and plot the cumulative returns from investing in this strategy from the start of 2006 to the end of 2014, re-investing the proceeds each period. Does the strategy generate a positive return on average?

- b. Based on the liquidity-adjusted CAPM presented in Chapter 10, we know that an asset's expected return should be adjusted for its covariance with the market return and its liquidity risk. Calculate the correlation between the returns of the carry trade strategy and the market portfolio (proxied by the S&P 500). Then calculate the correlation between the carry trade's returns and s_M (proxied by the average bid-ask spread for U.S. stocks). How will these two correlations impact the strategy's required return? Given your findings, does the strategy appear more or less attractive? In answering this question, you may ignore the currency mismatch between the carry trade strategy in CHF and the S&P 500 index, which is in USD.
- c. As illustrated by your answer to part a, the carry trade is generally profitable but suffered during the financial crisis in 2008. Based on your knowledge of the limits to arbitrage theory presented in Section 11.2, what could be a reason that more speculative capital did not flow into the carry trade during the 2008 crisis? Now plot the carry trade returns together with the TED spread. The TED-spread is the difference between the rate at which banks can borrow (USD Libor at the time) and the three month Treasury bill rate (a proxy for the USD risk-free rate). It can be used as a measure for the cost of raising speculative capital. Does the figure support your explanation?
- d. Imagine that you are raising capital for an investment vehicle targeting carry trades, however you are worried that investors will pull their money out during a crisis and you will not be able to raise new funds. Which of the following fund structures would alleviate such concerns:

1. A structure where investors can withdraw their money at any time.
2. Same as 1, but with a clause allowing the fund to freeze redemptions for up to a year during a market crisis.
3. A fund where investors' money is locked up for a fixed 10-year period.